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NetPulse — Network Monitoring & Health Dashboard



ACADEMIC PROJECT • B.TECH CSE

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NETPULSE | PROJECT OVERVIEW

Network monitoring checks whether devices can be reached and records what happened over time.

THE CHALLENGE

Checking many routers, servers and PCs by hand takes time.

LOST CONTEXT

Past checks are hard to compare when results are scattered.

NETPULSE

One dashboard shows device state, check history and alerts.

Project goal

Give a small IT team a clear view of configured device availability and recent health changes.

PROJECT OBJECTIVES & FEATURES

01	**Monitor devices**	Check enabled devices with ping.
02	**Show health**	Display status and measured latency.
03	**Keep history**	Save each monitoring result.
04	**Raise alerts**	Create alerts for status and latency changes.
05	**Keep an audit trail**	Record activity and alert events.

Implemented areas: dashboard • device management • alerts • logs • authentication

TECHNOLOGY STACK

Python

Application language

FastAPI + Pydantic

Routes + request validation

HTML • CSS • JS

Browser interface

Cookie sessions

Login, roles and CSRF

SQLAlchemy

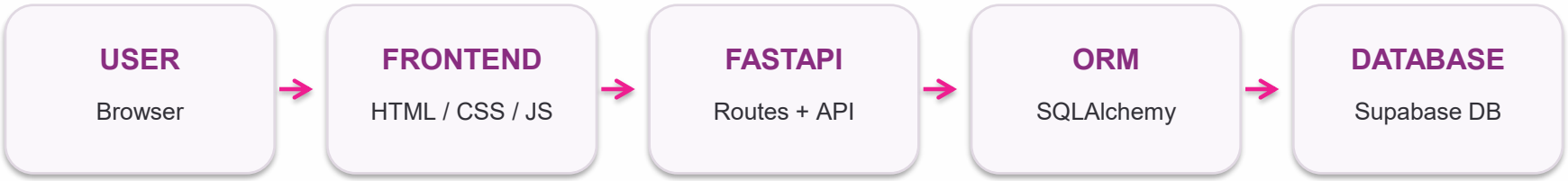
ORM and database queries

Supabase PostgreSQL

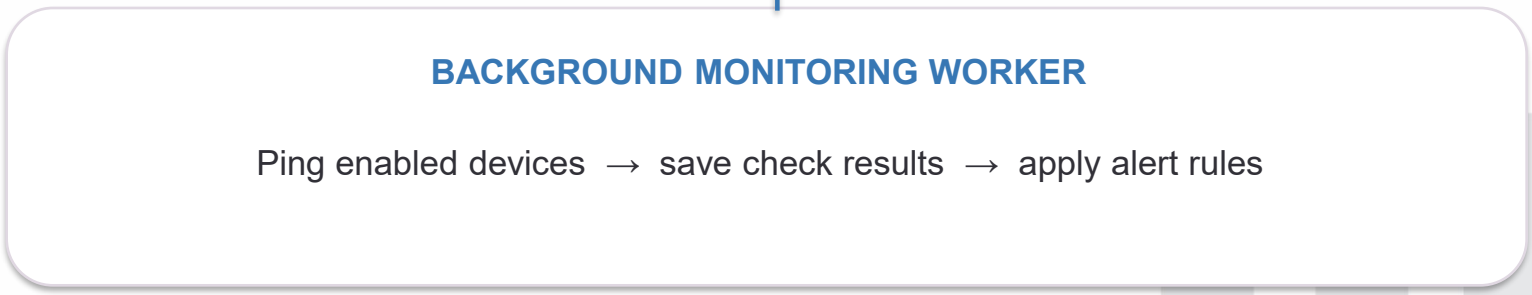
Hosted relational database

FastAPI serves pages and JSON APIs; database credentials stay on the server.

SYSTEM ARCHITECTURE



Request / response path



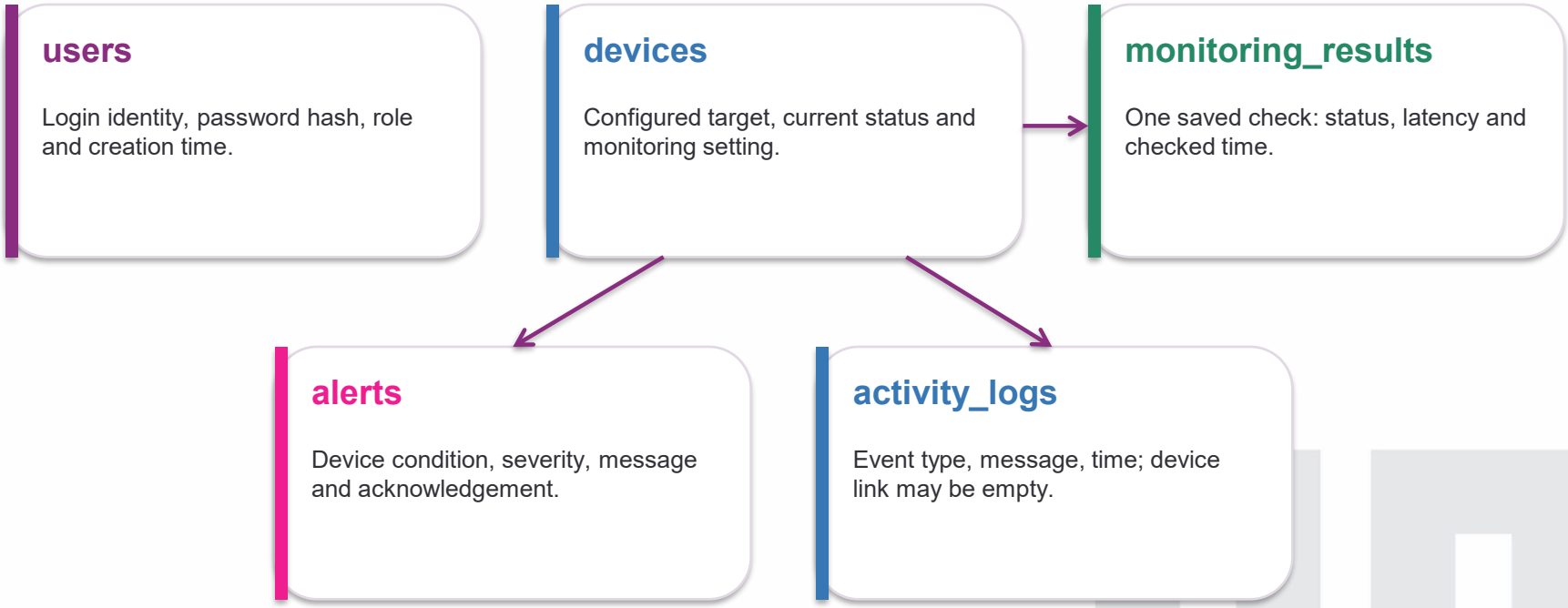
A background thread checks devices and uses its own SQLAlchemy session for each check.

BACKEND WORKING

1 REQUEST	Browser calls a FastAPI page or JSON API.
2 CHECK	Protected routes check the session; Pydantic validates inputs and CSRF protects writes.
3 PROCESS	A route reads and updates SQLAlchemy models through a database session.
4 RESPONSE	FastAPI returns Jinja HTML or JSON; JavaScript fetches and displays API data.

Examples: /api/dashboard/summary • /api/devices • /api/alerts

DATABASE DESIGN | VERIFIED MODELS



Verified links: devices → monitoring_results, alerts and activity_logs. Users has no model FK link.

NETWORK MONITORING WORKFLOW

01 Load enabled devices → **02** Send one OS ping → **03** Classify status + latency → **04** Save result and device state → **05** Apply transition alert rules

STATUS RULES

Ping succeeds → ONLINE, WARNING or CRITICAL based on latency thresholds. No response → OFFLINE.

ALERT RULES

Status transitions create down / recovery alerts; high latency can create alerts. Active duplicates are avoided.

Runs every configured MONITOR_INTERVAL seconds (default: 10); checks run from this host.

USER INTERFACE | ILLUSTRATIVE MOCKUPS

Illustrations based on implemented screens; these are not application screenshots.

DASHBOARD

DEVICES | ONLINE | ALERTS

STATUS SUMMARY • LATENCY HISTORY

RECENT ALERTS • ACTIVITY

DEVICE MANAGEMENT

Search devices Filter: status

Router • ONLINE

Server • WARNING

Switch • UNKNOWN

Other views: device details and history • alert filters • activity log search

BENEFITS & APPLICATIONS

A shared view helps teams spot reachability changes and review recent checks.

IT SUPPORT

Review device health and recent status changes from one place.

SYSTEM ADMINS

Track configured infrastructure and investigate recorded alerts.

LABS & TEAMS

Check availability of lab PCs, servers, routers and switches.

NetPulse reports device reachability; it does not repair failures or detect cyberattacks.

LIMITATIONS & FUTURE SCOPE

CURRENT SCOPE

- Uses OS ping for reachability
- Checks run from the application host
- Measures ping latency; not full device telemetry
- Alert delivery is in-app; no email integration

POSSIBLE NEXT STEPS

- Add email / notification integrations
- Expand uptime and historical analytics
- Add protocols such as SNMP
- Review role and access options
- Improve deployment for larger setups

Future items are proposals; they are not claimed as implemented.

CONCLUSION & Q&A

NetPulse turns repeated device checks into a clear, searchable health history.

MONITOR

Ping configured devices and classify health.

STORE

Use SQLAlchemy with Supabase PostgreSQL.

REVIEW

See status, alerts and activity in FastAPI pages.

THANK YOU

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